Chapter 4:
Sequences

4.2 Sequences

Definition 4.6. *A* <u>*sequence*</u> f *is a function* $f : \mathbb{N}^+ \to \mathbb{R}$.

Note that we often denote a sequence as

$$(f_n) = (\, f_1, \;\; f_2, \;\; f_3, \;\; f_4, \dots \,)$$

rather than

$$f(1), \;\; f(2), \;\; f(3), \;\; f(4), \dots$$

Example. $f: \mathbb{N}^+ \to \mathbb{R}$ defined by $f(n) = n^2 + 1$

Example. We can approximate π by the sequence

$$(f_n) = (\, 3, \;\; 3.1, \;\; 3.14, \;\; 3.141, \;\; 3.1416, \;\; 3.14159, \dots \,)$$

Example. The sequence

$$(-1,\ \ 1,\ \ -1,\ \ 1,\ \ -1,\ \ 1, \ldots)$$

does not converge, nor does

$$(2,\ \ 4,\ \ 6,\ \ 8,\ \ 10,\ \ 12, \ldots)$$

Example. The sequence

$$(f_n) = \left(4 + (-1)^n \frac{2}{\sqrt{n}}\right) = \left(4 - \frac{2}{\sqrt{1}}, 4 + \frac{2}{\sqrt{2}}, 4 - \frac{2}{\sqrt{3}}, 4 + \frac{2}{\sqrt{4}}, 4 - \frac{2}{\sqrt{5}}\right)$$

Converting these terms to rounded decimals

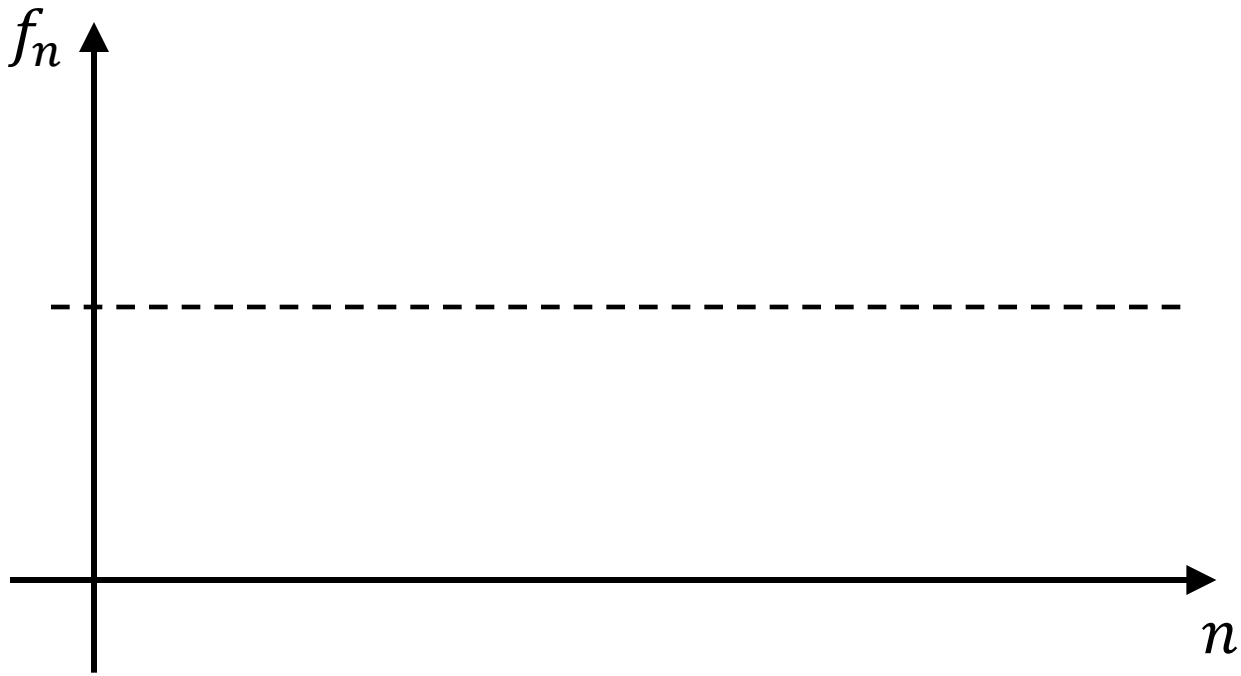
$$(f_n) = (\ 2,\ \ 5.414,\ \ 2.845,\ \ 5,\ \ 3.106,\ \ 4.816,\ \ 3.244,\ \ 4.707, \ldots)$$

Computing more terms

$$f_{100} = 4 + \frac{2}{\sqrt{100}} \approx 4.2 \qquad f_{1000} = 4 + \frac{2}{\sqrt{1000}} \approx 4.0632$$

$$f_{10000} = 4 + \frac{2}{\sqrt{10000}} \approx 4.02 \qquad f_{100000} = 4 + \frac{2}{\sqrt{100000}} \approx 4.0063$$

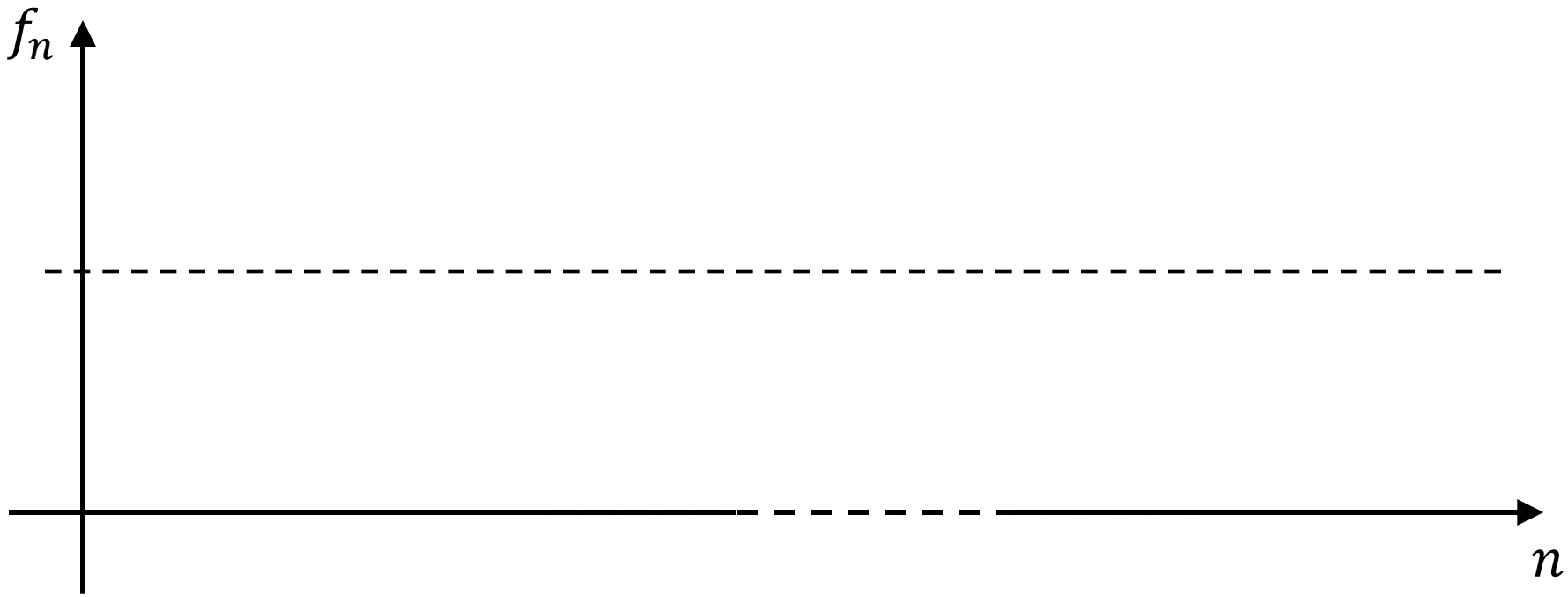
$$f_{100000} = 4 + \frac{2}{\sqrt{100000}} \approx 4.002$$



We expect to say that $\lim_{n\to\infty} f_n = 4$ if

For instance, when $n > 1000$, $|f_n - 4| < 0.0632$,

when $n > 10000$, $|f_n - 4| < 0.02$, etc.



Let $p(n, \epsilon)$ be the condition $|f_n - 4| < \epsilon$.

For example, consider $\epsilon = 0.0001$:

For arbitrary $\epsilon > 0$:

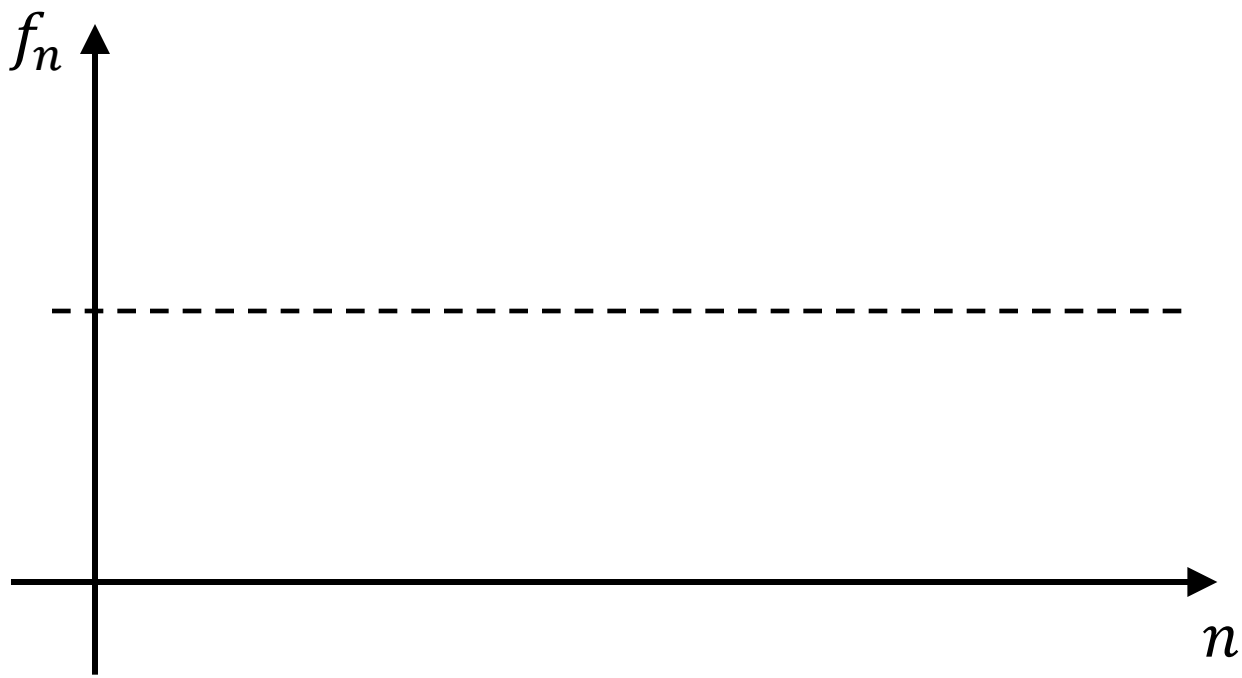
Definition 4.7. *Let (f_n) be a sequence and $L \in \mathbb{R}$.*

We say that (f_n) <u>converges</u> to L when:

For every $\epsilon > 0$ there exists $M \in \mathbb{N}^+$ s.t. for all $n \in \mathbb{N}^+$ if $n > M$ then $|f_n - L| < \epsilon$.

We write

$$f_n \to L \text{ or } \lim_{n\to\infty} f_n = L$$



Example. Let $f_n = 1 - 1/n$. Let's show that $f_n \to 1$.

Want: $\forall \epsilon > 0 \ \exists M \in \mathbb{N}^+$ s.t. $\forall n \in \mathbb{N}^+$ if $n > M$ then $|f_n - 1| < \epsilon$.

Scrap work. Take any $\epsilon > 0$ and find M:

Proof.

Example. Let $f_n = \frac{1}{n^2+1}$. Let's show that $f_n \to 0$.

Want: $\forall \epsilon > 0 \ \exists M \in \mathbb{N}^+$ s.t. $\forall n \in \mathbb{N}^+$ if $n > M$ then $|f_n - 0| < \epsilon$.

Scrap work. Take any $\epsilon > 0$ and find M:

Proof.

Definition 4.8. *Let* (f_n) *be a sequence. We say* (f_n) <u>*diverges*</u> *when the statement*

$$(f_n) \text{ converges to } L$$

Is false for every $L \in \mathbb{R}$. *In other words, when*

Example. The sequence

$$(-1,\ \ 1,\ \ -1,\ \ 1,\ \ -1,\ \ 1, \ldots)$$

diverges.

Proof.

Theorem 4.9. (Algebra of Limits Theorem)

Let (f_n) *and* (g_n) *be convergent sequence with* $f_n \to \alpha$ *and* $g_n \to \beta$. *Then*

i. $f_n + g_n \to \alpha + \beta$

ii. $f_n - g_n \to \alpha - \beta$

iii. $f_n g_n \to \alpha\beta$

iv. $f_n / g_n \to \alpha / \beta$ (*provided* $g_n \neq 0$ *for all* $n \in \mathbb{N}^+$ *and* $\beta \neq 0$)

Proof of i.

Theorem. *Let* (f_n) *and* (g_n) *be convergent sequences.*

- *If* $f_n \leq g_n$ *for all* $n \in \mathbb{N}^+$*, then*

$$\lim_{n\to\infty} f_n \leq \lim_{n\to\infty} g_n \,.$$

- *If* $f_n \leq b$ *for all* $n \in \mathbb{N}^+$*, then*

$$\lim_{n\to\infty} f_n \leq b.$$

- *If* $a \leq f_n$ *for all* $n \in \mathbb{N}^+$*, then*

$$a \leq \lim_{n\to\infty} f_n \,.$$

Example. Consider the following sequences

i. $(1, 2, 3, 4, 5, \dots)$

ii. $(-1, -2, -3, -4, -5, \dots)$

iii. $(1, 4, 9, 16, 49, \dots)$

iv. $(-2, -4, -8, -16, -32, \dots)$

We expect them to diverge since they are *unbounded*.

That is:

Theorem 4.11. *Let* (f_n) *be a sequence. If* (f_n) *is not bounded, then* (f_n) *does not converge.*

Definition 4.10. Let (f_n) be a sequence. We say that (f_n) is <u>bounded</u> if the set

$$\{ f_n \mid n \in \mathbb{N}^+ \}$$

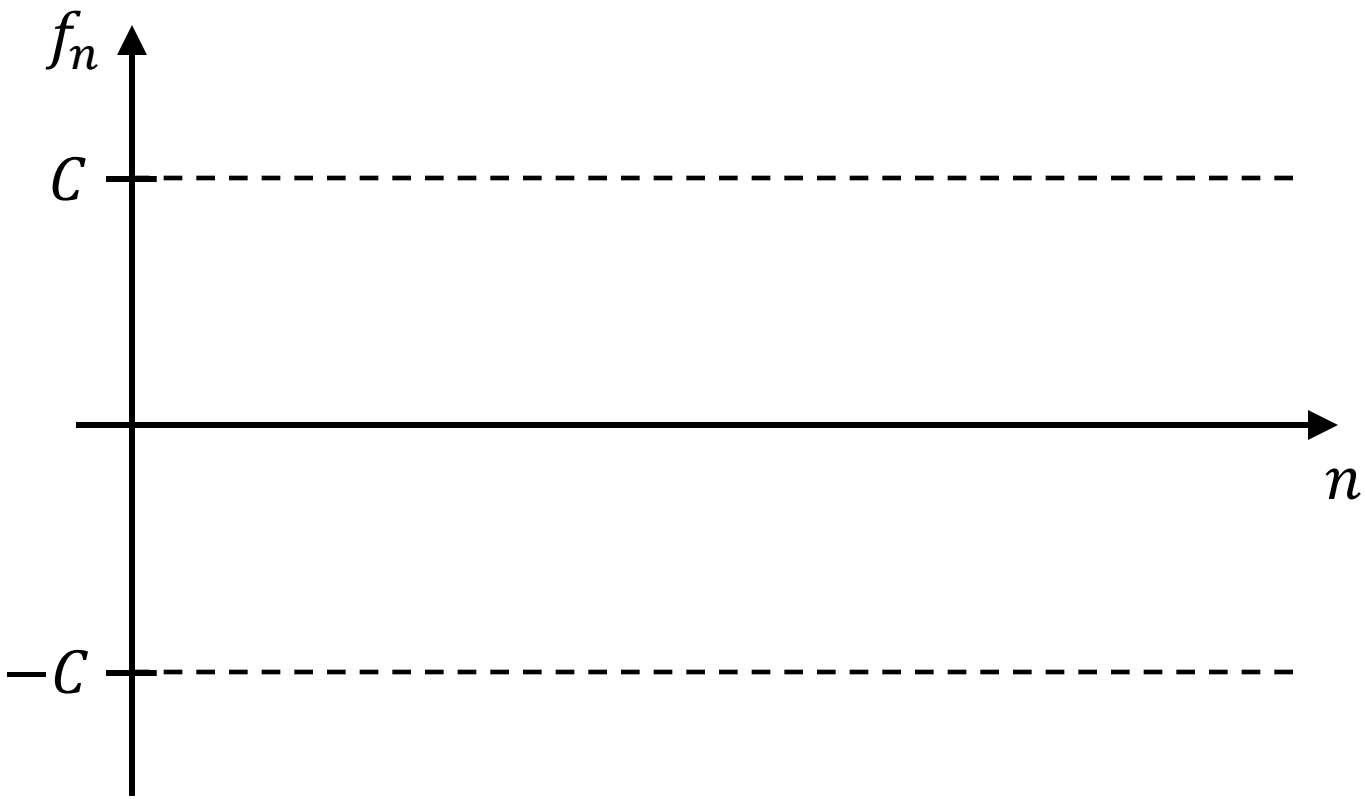
is bounded both above and below.

If (f_n) is not bounded, we say (f_n) is <u>unbounded</u>.

Notice that (f_n) is bounded means that

(f_n) is bounded when

$$-C \leq f_n \leq C \text{ for all } n \in \mathbb{N}^+.$$



Theorem 4.12. *A sequence* (f_n) *is bounded if and only if there exists* $C > 0$ *so that*

$$|f_n| \leq C \text{ for all } n \in \mathbb{N}^+.$$

Now let's state and prove the contrapositive of Theorem 4.11:

Theorem 4.13. *Let* (f_n) *be a sequence. If* (f_n) *converges, then* (f_n) *is bounded.*

Example. The following sequence is bounded but divergent

$$(-1,1,-1,1,\dots)$$

as it is an interleaving of two different sequences

Definition 4.16. *Let* (f_n) *be a sequence and let* $n_1, n_2, n_3, \ldots$ *be integers such that*

$$0 < n_1 < n_2 < n_3 < \cdots$$

We say that

$$(f_{n_1}, f_{n_2}, f_{n_3}, \ldots)$$

is a <u>*subsequence*</u> *of* (f_n). *We denote a subsequence of* (f_n) *as* (f_{n_k}).

Theorem 4.14. *A sequence* (f_n) *converges to* L *if and only if every subsequence of* (f_n) *converges to* L.

Proof.

Theorem 4.14. *A sequence* (f_n) *converges to* L *if and only if every subsequence of* (f_n) *converges to* L.

Notice that Theorem 4.14 contains:

If (f_n) *converges to* L, *then every subsequence of* (f_n) *converges to* L.

The contrapositive of this statement is:

If at least one subsequence of (f_n) *does not converge to* L,
then (f_n) *does not converge to* L.

It follows that:

Corollary 4.15. *Let* (f_n) *be a sequence. If* (f_n) *has two subsequences each of which converge to a different value, then* (f_n) *diverges.*

Corollary 4.17. *Let* (f_n) *be a sequence. If* (f_n) *has a subsequence which diverges, then* (f_n) *diverges.*

Theorem 4.18 (Divergence Criteria Theorem)

Let (f_n) be a sequence. If any of the following are true, then (f_n) diverges:

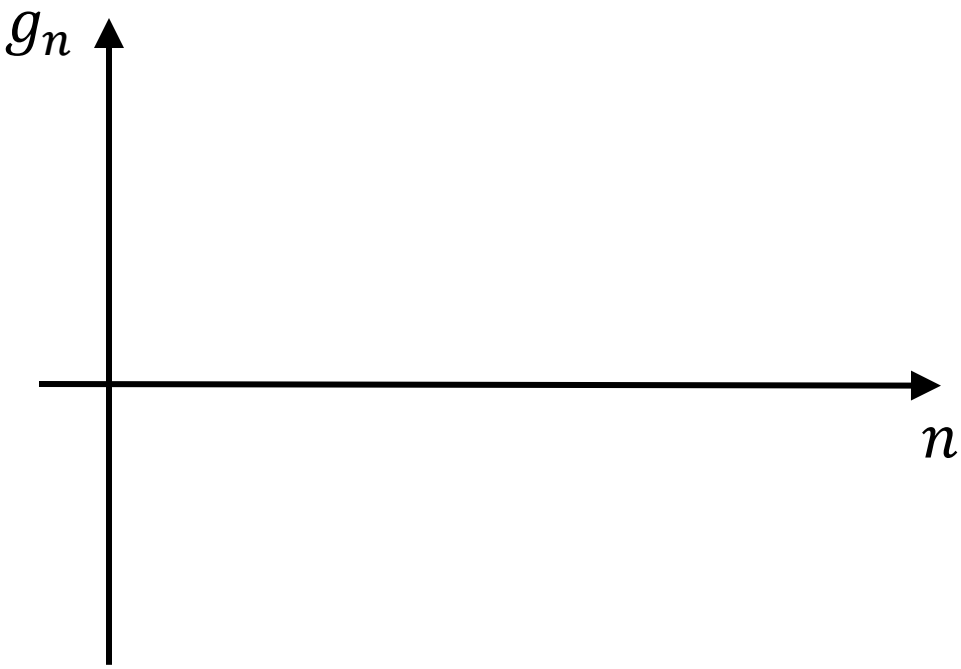
1. (f_n) *is unbounded;*
2. (f_n) *has two subsequences each of which converge to a different value;*
3. (f_n) *has a subsequence that diverges.*

Recall that:

Theorem 4.13. *Let* (f_n) *be a sequence. If* (f_n) *converges, then* (f_n) *is bounded.*

Examples

- $f_n = (-1)^n$ is bounded but divergent
- $g_n = 1 - 1/n$ is bounded and convergent



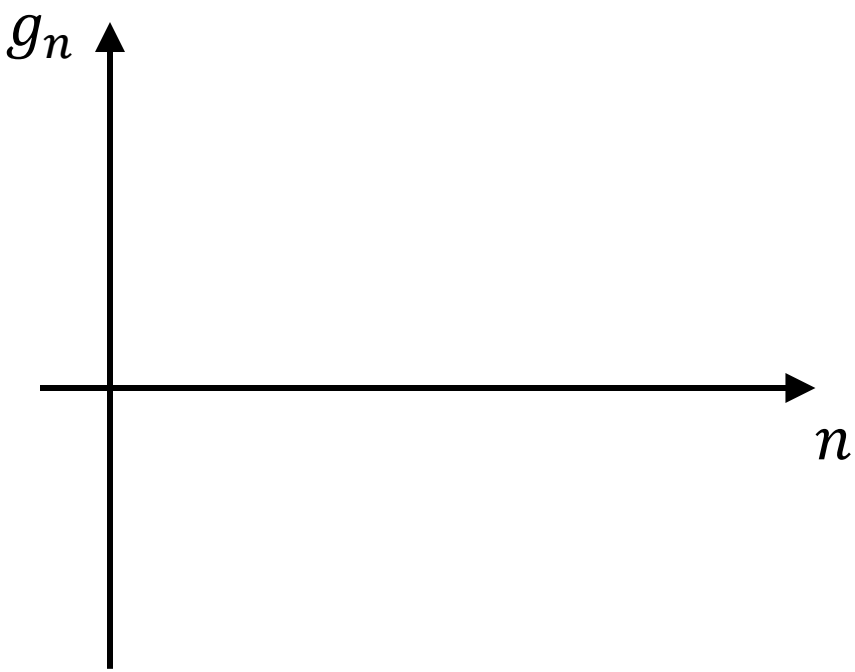
Definition 4.21. *Let (f_n) be a sequence.*

We say that (f_n) is <u>monotone increasing</u> when $f_n \leq f_{n+1}$ for all $n \in \mathbb{N}^+$.

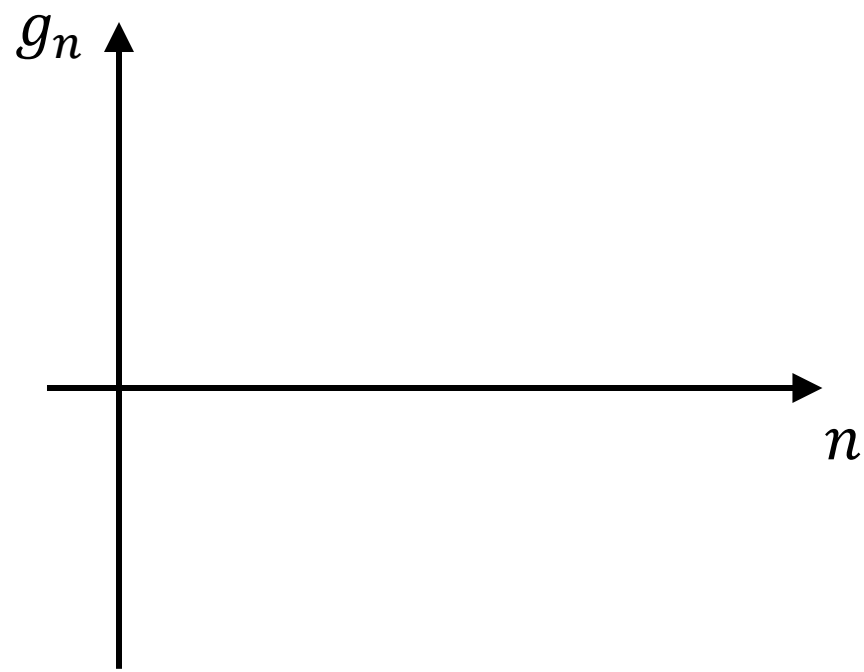
We say that (f_n) is <u>monotone decreasing</u> when $f_n \geq f_{n+1}$ for all $n \in \mathbb{N}^+$.

We say that (f_n) is monotone when (f_n) is either monotone increasing or monotone decreasing.

Increasing



Decreasing



Lemma 4.23. *Let* (f_n) *be a bounded sequence. If* (f_n) *is monotone increasing, then* (f_n) *converges.*

Theorem 4.25. *Let* (f_n) *be a monotone sequence. The sequence* (f_n) *converges if and only if* (f_n) *is bounded.*

Corollary 4.26. *Let* (f_n) *be a bounded sequence and let* $A_f = \{f_n \mid n \in \mathbb{N}^+\}$.

- *If* (f_n) *is monotone increasing, then* $f_n \to \sup A_f$.
- *If* (f_n) *is monotone decreasing, then* $f_n \to \inf A_f$.

Notice that for a bounded increasing monotone sequence (f_n), as n gets larger the terms f_n get squeezed closer and closer to $\sup A_f$.

Hence they get squeezed closer and closer to each other!

In other words:

Definition 4.27. *Let* (f_n) *be a sequence. We say* (f_n) *is* <u>*Cauchy*</u> (*or is a* <u>*Cauchy sequence*</u>) *when:*

For every $\epsilon > 0$ *there exists* $M \in \mathbb{N}^+$ *such that if* $j, k > M$ *then* $\left|f_j - f_k\right| < \epsilon$.

Example 4.28. Prove $(f_n) = \left(\frac{(-1)^n}{n}\right)$ is Cauchy.

Theorem 4.32 (Cauchy Convergence Criterion)

Let (f_n) *be a sequence. The sequence* (f_n) *converges if and only if* (f_n) *is Cauchy*.

To prove this theorem, we need to prove:

1. If (f_n) converges, then (f_n) is Cauchy.
2. If (f_n) is Cauchy, then (f_n) converges.

Theorem 4.29 *Let* (f_n) *be a sequence. If* (f_n) *converges then* (f_n) *is Cauchy*.

Theorem. *Let* (f_n) *be a sequence. If* (f_n) *is Cauchy then* (f_n) *converges*.

To prove this theorem, we need to prove:

1. Bolzano-Weierstrass Theorem. If (f_n) is a bounded sequence, then (f_n) has a convergent subsequence.

2. If (f_n) is a Cauchy sequence, then (f_n) is bounded.

3. If (f_n) is a Cauchy sequence and (f_n) has at least one convergent subsequence, then (f_n) is convergent.

Theorem. *Let (f_n) be a sequence. If (f_n) is Cauchy then (f_n) converges.*

To prove this theorem, we need to prove:

1. Bolzano-Weierstrass Theorem. If (f_n) is a bounded sequence, then (f_n) has a convergent subsequence.

2. If (f_n) is a Cauchy sequence, then (f_n) is bounded.

   Idea of proof. Argue as in Theorem 4.13 (convergent sequences are bounded) that for $\epsilon = 1$ there exists $M \in \mathbb{N}^+$ such that if $j, k > M$ then $\left|f_j - f_k\right| < 1$.

   Hence if $k > M$ then $|f_k| < |f_{M+1}| + 1$.

3. If (f_n) is a Cauchy sequence and (f_n) has at least one convergent subsequence, then (f_n) is convergent.

Bolzano-Weierstrass Theorem. *If* (f_n) *be a bounded sequence, then* (f_n) *has a convergent subsequence*.

Idea of proof: Show that (f_n) contains a monotone subsequence (f_{n_k}). Since (f_{n_k}) is also bounded, then (f_{n_k}) converges.

Definition. (f_n) *has a* <u>*peak*</u> *at* $n = s$ *if* $n > s \Rightarrow f_n \leq f_s$.

Case 1. (f_n) *has infinitely many peaks at* $n = n_k$ *for* $n_1 < n_2 < n_3 < \cdots$.

Bolzano-Weierstrass Theorem. *If* (f_n) *be a bounded sequence, then* (f_n) *has a convergent subsequence*.

Case 2. (f_n) *has finitely many peaks*.

Theorem. *If* (f_n) *is a Cauchy sequence and* (f_n) *has at least one convergent subsequence, then* (f_n) *converges*.

Definition 4.33. *A sequence* (f_n) *is called contractive if there exists a constant* c *with* $0 < c < 1$ *such that for all* $n \in \mathbb{N}^+$

$$|f_{n+2} - f_{n+1}| \leq c|f_{n+1} - f_n|$$

Theorem 4.34. *If* (f_n) *is a contractive, then* (f_n) *converges.*

Example. Consider the polynomial $f(x) = x^7 - 5x + 1$. From plotting the graph of $y = f(x)$ we can guess that this polynomial has three real roots, and there is a unique root r of $f(x)$ with $0 < r < 1/2$.

4.3 Sequences of Functions

So far, we have only studied sequences of real numbers.

We can also study sequences of other mathematical objects, such as functions.

Example. For all $n \in \mathbb{N}^+$ let $g_n(x)$ denote the following function (of n and x):

$$g_n(x) = \frac{2x^2 + 2nx}{n}$$

For each particular value of n we produce a function $g_n: \mathbb{R} \to \mathbb{R}$.

Example. For all $n \in \mathbb{N}^+$ let $g_n(x)$ denote the following function (of n and x):

$$g_n(x) = \frac{2x^2 + 2nx}{n}$$

For each particular value of x in the domain, we produce a sequence of real numbers $(g_n(x))$.

Definition 4.33. *Let* $D \subseteq \mathbb{R}$. *Let* $(f_n(x))$ *be a sequence of functions* $f_n : D \to \mathbb{R}$ *and let* $f : D \to \mathbb{R}$ *be a function.*

We say that $(f_n(x))$ <u>*converges pointwise*</u> *to* $f(x)$ *when*

$$\textit{for every } x \in D \textit{ we have } f_n(x) \to f(x).$$

In other words

$$\forall x \in D\ \forall \epsilon > 0\ \exists M = M(x, \epsilon) \in \mathbb{N}^+ \textit{ s.t. if } n > M \textit{ then } |f_n(x) - f(x)| < \epsilon.$$

Now recall our definition of convergence of a sequence (f_n):

Definition. *We say that a sequence* (f_n) <u>*converges*</u> *to* L *when*

$$\forall \epsilon > 0 \; \exists M \in \mathbb{N}^+ \textit{ such that if } n > M \textit{ then } |f_n - L| < \epsilon.$$

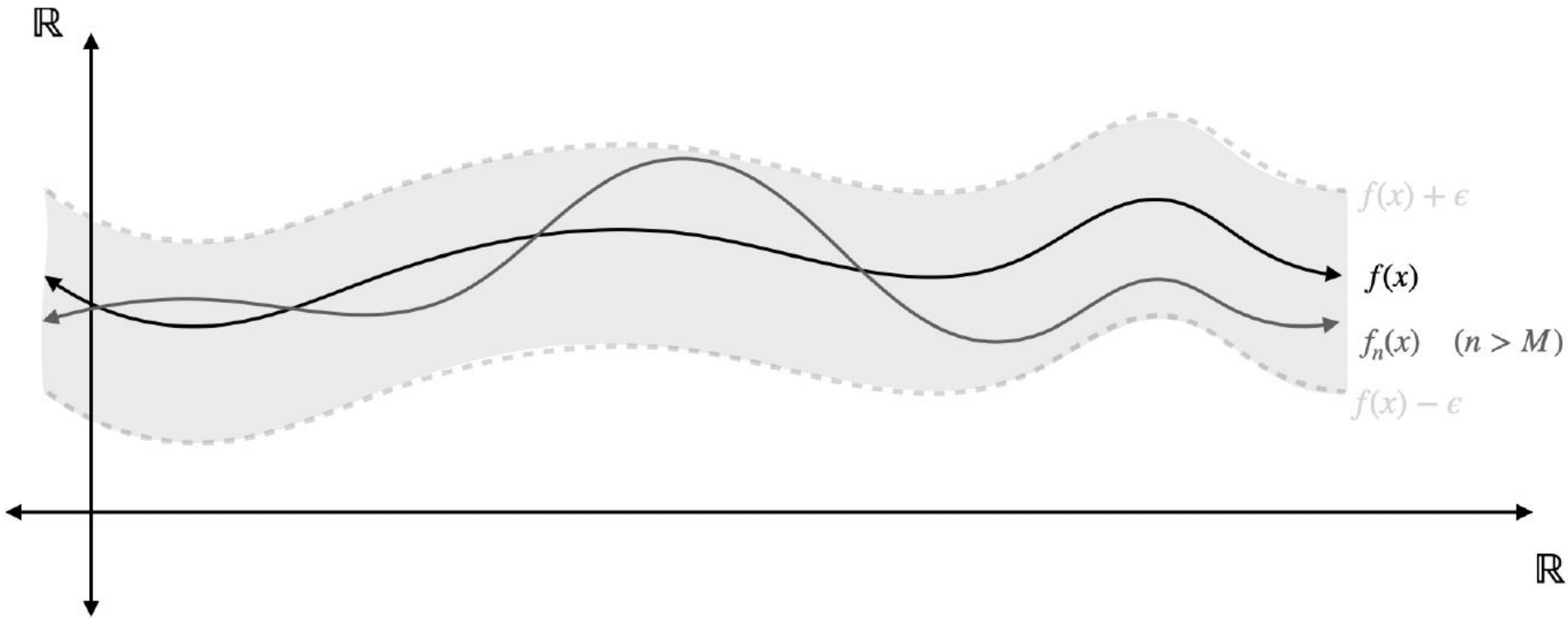
In other words, f_n estimates L with an error $< \epsilon$.

Unfortunately, for sequences of functions, $f_n(x)$ can converge to $f(x)$ at different rates for different values of x.

Definition 4.33. *Let $D \subseteq \mathbb{R}$. Let $(f_n(x))$ be a sequence of functions $f_n : D \to \mathbb{R}$ and let $f : D \to \mathbb{R}$ be a function.*

We say that $(f_n(x))$ <u>converges uniformly</u> to $f(x)$ when

$$\forall \epsilon > 0 \; \exists M = M(\epsilon) \in \mathbb{N}^+ \; s.t. \; \forall x \in D \; if \; n > M \; then \; |f_n(x) - f(x)| < \epsilon.$$



Example. For all $n \in \mathbb{N}^+$ let $g_n(x)$ denote the following function (of n and x):

$$g_n(x) = \frac{2x^2 + 2nx}{n}$$

Show that $(g_n(x))$ does not converge uniformly to $2x$ for $x \in \mathbb{R}$.